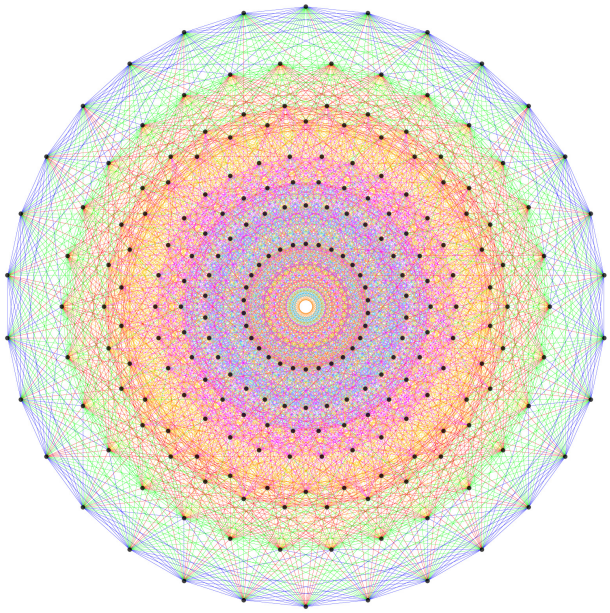
Representation Theory

Maschke theorem: Brief Exposition

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Introduction

This exposition aims at exploring the basic foundations of representation theory of finite groups. The main outcome is to explore Maschke's theorem that provides a sufficient condition for when a group algebra is semisimple. We also explore Artin-Wedderburn theorem that characterizes the structure of the semisimple algebras irreducible and finite dimensional representations. The exposition assumes basic knowledge in group, ring and field theory and linear algebra

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Chapter 1

Linear Algebra

We begin by defining what is an Algebra. This is the main structure we will be using throughout this paper. To give you an intuition first, an Algebra is basically a set with a structure of both a ring an a vector space.

1.1 Algebras over a Field

Definition 1.1 (k-Algebra)

Let k be any field. We say that a $k-$vector space A is a k-algebra if A is equipped with an associative bilinear operation (product)

$$\times : A \times A \to A$$

satisfying the following axioms of bilinearity. For every $x, y, z \in A$ and $a, b \in k$:

1. $z \times (x + y) = z \times x + z \times y$
2. $(x + y) \times z = x \times z + y \times z$
3. $(a \cdot x) \times (b \cdot y) = ab \cdot (x \times y)$

where $\cdot$ is the vector space scalar multiplication.

Another way to define a k-algebra is to first define it as a ring equipped with scalar multiplication operation by embedding the field k inside A.

Definition 1.2 (k-Algerba)

Let k be a field. A $k-$algebra A is a ring with an injective ring homomorphism $k \to A$, where for any $\lambda \in k$ and $a \in A$ we have that

$$\lambda \cdot a = a \cdot \lambda$$

i.e. multiplication by elements in the image of k in A is commutative.

Note. 1. Observe that we can define the vector space structure on A in the definition above by defining the scalar product λa to be the product of the image of λ in A times a to get $\lambda \cdot a$ for all $\lambda \in k$ and $a \in A$. That is

$$\lambda a = \lambda \cdot a$$

2. An algebra product operation might not be commutative, just like a ring need not be commutative. If so we just say it is a commutative algebra.

3. We may or may not include 1 as a unity in our definition of an algebra but here we do. And the operation is associative.

Definition 1.3 (k-Algebra Homomorphism)

Let A, B be $k-$algebras. A function $\varphi : A \to B$ is a $k-$algebra homomorphism if it is a linear map of vector spaces and for any $a \in A$ and $b \in B$ we have that

$$\varphi(ab) = \varphi(a)\varphi(b)$$

1.2 Direct Sum and Direct Product

Now, we get to differentiate between two important notions in linear algebra: Internal direct sum and external direct product. Sometimes for simplicity we just say Direct Sum and Direct Product.

Definition 1.4 (Internal Direct Sum)

Let V be a vector space. We say V is the internal direct sum of two subspaces $U, W \subseteq V$ denoted by $V = U \oplus V$ iff every vector $v \in V$ could be written uniquely as a sum of two vectors u, w where $u \in U$, and $w \in W$.

That is, if

$$V = U + W.$$

and V is the internal direct sum of U and V, then there are unique vectors $u \in U$ and $w \in W$ s.t.

$$v = u + w.$$

The above definition tells us exactly when a sum of two subspaces $U + W$ of a given vector space V is a direct sum, and in this case we call V an internal direct sum since the two spaces U and W are subspaces. Next, we define an external direct product of two separate given spaces.

Definition 1.5 (External Direct Product)

Let K, L be two vector spaces. We define the *External Direct Product* of K and L to be the vector space

$$K \times L := \{(k, l) \mid k \in K,\ l \in L\}$$

with operations of component wise addition and scalar multiplication.

The next theorem provides a complete characterization for when an external direct product of two subspaces and their sum are exactly the same, giving also an equivalent

condition for when a sum is an internal direct sum.

Theorem 1.6

Let U, W be subspaces of a vector space V. **TFAE**

1. The map
$$\pi : U \times W \to U + W$$
given by
$$(u, w) \mapsto u + w$$
is an isomorphism.

2. $U \cap W = \{0\}$.

3. Any vector $x \in U + W$ could be written uniquely in the form
$$u + w$$
with $u \in U$ and $w \in W$.

Proof. (1) $\Rightarrow$ (2) Suppose for the sake of contradiction that there exist a non-zero vector
$$v \in U \cap W.$$
So $v \in U$ and $v \in W$. Hence via the isomorphism π, the element $(v, 0) \in U \times W$ maps to $v + 0 = v$ and $(v, 0) \in U \times W$ maps to $0 + v = v$ violating injectivity of π.

(2) $\Rightarrow$ (3) Suppose that there is $x \in U + W$ such that x has two different representations
$$x = a + b = c + d.$$

where $a, c \in U$ and $b, d \in W$ Then

$$a + b = c + d$$

$$\Rightarrow a - c = d - b.$$

Since U, W are subspaces that is closed under addition we get that $a - c \in U$ and $b - d \in W$. Since $a - c = d - b$, therefore,

$$a - c \in U \cap W.$$

which implies that $a - c = d - b = 0$. Hence, $a = c$ and $b = d$ as desired.

(3) $\Rightarrow$ (1) Since any vector $x \in U + W$ could be written as $u + w$, where $u \in U$ and $w \in W$ this implies that π is surjective as a preimage for any element $x = u + w$ is (u, w). By uniqueness of this representation of elements in $U + W$ we get that the preimage of an element $x = u + w$ is unique. ■

So, if we have a vector space that is a sum of two subspaces

$$U + W = V$$

and satisfies one of the conditions above we say V is an internal direct sum of U, W and we write

$$V = U \oplus W.$$

Moreover, we can see that the two notions of internal direct sum and external direct product coincide when K, L are subspaces of some vector space V and V is an external direct sum of K and L as proved the direct sum & product are isomorphic.

Definition 1.7 (Direct sum of Algebras)

Let A, B be two $k-$algebras. We define the direct sum $A \oplus B$ to be the $k-$algebra whose elements are pairs (a, b) where $a \in A$ and $b \in B$, with component wise operations.

Chapter 2

Basics of Representation Theory

2.1 Introduction and Motivation

The main goal of representation theory of finite groups is to understand abstract algebraic structures—specifically groups and associative algebras—by concretely realizing their elements as linear transformations of vector spaces. By transforming non-linear group multiplication into the well-understood domain of linear algebra and matrix arithmetic, one converts difficult group-theoretic questions into tractable eigenvalue problems, trace computations, and spectral decompositions.

First we define what is a representation? then we move to defining a homomorphism and isomorphism of representations. Moreover, we introduce the definition of an important structure that we will be using extensively throughout this exposition which is the group algebra $k[G]$, then we will move to working with simple representations/modules. In between we show how can modules and representations be seen as the same thing.

2.2 Basic Definitions

Definition 2.1 (Representation of an Algebra)

Let A be a $k-$algebra, and V be a $k-$vector space. A *representation* of A is a pair (V, ρ) where $\rho : A \to \text{End}(V)$ is a $k-$algebra homomorphism.

Definition 2.2 (Subrepresentation)

Let (V, ρ) be a representation of any $k-$algebra A. We say that a subspace $W \subseteq V$ is a subrepresentation iff elements of W commute with the representation ρ or that W is invariant under the action of A. That is for any $a \in A$ and any $w \in W$ we have that

$$\rho(a) \cdot w \in W$$

Definition 2.3 (Intertwining Operator)

Let (V, ρ_1) and (W, ρ_2) be any two representations of a $k-$algebra A, an intertwining operator $T : V \to W$ is a linear map that commutes with the action of A, that is

$$T(\rho_1(a)(v)) = \rho_2(a)(T(v))$$

for all $a \in A$ and $v \in V$. We may also say that T is a homomorphism of representations. If T is invertible, then it is an isomorphism of representations

Now, we show that the inverse of an intertwining operator is also intertwining.

Proof. Let V, W be as above. Let $T : V \to W$ be an invertible intertwining operator and $T^{-1} : W \to V$ be its inverse. Pick any $w \in W$, and consider its unique preimage $v \in V$. So,

$$w = T(v) \quad \text{and} \quad v = T^{-1}(w)$$

By definition of an intertwining operator we get:

$$T(\rho_1(a)(v)) = \rho_2(a)(T(v)) \iff T(\rho_1(a)(T^{-1}(w))) = \rho_2(a)(w)$$

Applying T^{-1} on both sides and rearranging the terms:

$$T^{-1}(\rho_2(a)(w)) = \rho_1(a)(T^{-1}(w))$$

for any $w \in W$. Hence, T^{-1} is an intertwining operator as desired. ■

2.2.1 Equivalence of Representations and Modules

Now, how are representations of any $k-$algebra A could be seen as A-modules?

We basically define the action of A on V by the action of the image of A in End(V) as linear transformations. For the other direction, Let M be an $A-$module. Then M is an abelian group. We define scalar multiplication via the action of A, as we mentioned earlier A has a copy of the field k inside it, hence: for any $\lambda \in k$ and any $m \in M$ we have that

$$\lambda \cdot m = \lambda m$$

w where $\cdot$ is the action of A on M. It remains to prove that the action of A on M as a module defines group algebra homomorphism from A to End(M).

Proof. Fix any $a \in A$, consider the map

$$\varphi(a) : M \to M$$

given by $m \mapsto a \cdot m$. By the axioms of a module action we get

- $\varphi(a)(m+n) = a \cdot (m+n) = a \cdot m + a \cdot n$
- $\varphi(a)(\lambda m) = a \cdot (\lambda m) = (a\lambda) \cdot m = (\lambda a) \cdot (m) = \lambda(\varphi(a)(m))$

the last equality is because we defined the image of the field k in A to be central. So, we proved that $\varphi(a)$ is a linear map for any $a \in A$. Now, we define the homomorphism of algebras

$$\rho : A \to \mathrm{End}(M)$$

given by $a \mapsto \varphi(a)$. To see that it is an algebra homomorphism, first we use the linearity of $\varphi(a)$

- $\rho(a+b) = \varphi(a+b) = \varphi(a) + \varphi(b) = \rho(a) + \rho(b)$
- $\rho(\lambda a) = \varphi(\lambda a) = \lambda\varphi(a) = \lambda\rho(a)$

since M is an A module, then by associativity of the action and the definition of $\varphi(a)$ for any $a \in A$, we get that

- $\rho(ab) = \varphi(ab) = \varphi(a)\varphi(b) = \rho(a)\rho(b)$

■

2.3 Representation of a Group

We are done discussing the general case. But what about the definition of a group representation? There are two ways to view this definition. One way is to see it, which I find really interesting, is to see it as an extension of a group action.

First, recall the definition of a group action on an arbitrary set S.

Definition 2.4 (Group Action)

Let G be a multiplicative group, and S to be any set. We define a group action of G on S to be a function

$$\cdot : G \times S \to S$$

denoted by $g \cdot x$ for all $g \in G$ and $x \in S$, satisfying the following axioms:

1. $e \cdot x = x \quad \forall\, x \in S$.

2. $g \cdot (h \cdot x) = gh \cdot x \quad \forall g, h \in G$ and $\forall x \in S$.

Using the previous two axioms of a group action, one can see that it actually induces a group homomorphism $\alpha : G \to \mathrm{Sym}(S)$, where $\mathrm{Sym}(S)$ is the group of all permutations on S with function composition.

Proof. Let $\cdot : G \times S \to S$ be a group action. Consider the map

$$\alpha : G \to \mathrm{Sym}(G)$$

where $\alpha(g)$ is a bijection on S defined by

$$\alpha(g)(x) = g \cdot x$$

for all $g \in G$ and $x \in S$. We claim that α is a group homomorphism. By the axioms of the group action, for any $g, h \in G$:

$$\alpha(gh)(x) = gh \cdot x = g \cdot (h \cdot x) = \alpha(g)(\alpha(h)(x)) = \alpha(g) \circ \alpha(h)(x)$$

■

Now, we repeat the same process but using a group action of a group G on a vector space V instead of an arbitrary set S requiring an additional property for the group action which is to be *linear*.

Definition 2.5 (Linear Group Action)

Let V be a $k-$vector space, and G be any group. Define the group action of G on V as the function

$$\cdot : G \times V \to V$$

that satisfies the following axioms for all $v, w \in V$, for all $g, h \in G$ and all $\lambda \in k$:

1. $e \cdot v = v$
2. $g \cdot (h \cdot v) = gh \cdot v$
3. $g \cdot (v + w) = g \cdot v + g \cdot w$
4. $g \cdot (\lambda v) = \lambda(g \cdot v)$

This definition is in fact equivalent to having a group homomorphism $\varphi : G \to \mathrm{GL}(V)$, exactly in the same way a regular group action gives rise to a homomorphism of groups from G to the symmetric group on an arbitrary set S. This pair of the homomorphism φ along with the vector space V are called a *representation* of the group G.

Definition 2.6 (Representation of a Group)

Let G be a group, and let V be a vector space over a field k. A *representation* of G is a pair (V, φ) where $\varphi : G \to \mathrm{GL}(V)$ is a group homomorphism given by the linear action of G on V.

Definition 2.7 (G-Homomorphism)

Let (ρ_1, V_1) and (ρ_2, V_2) be two representations of a group G over k. A linear map $T \colon V_1 \to V_2$ is called an *intertwining operator* (or *G-homomorphism*) if it commutes with the G-action:

$$T \circ \rho_1(g) = \rho_2(g) \circ T \quad \text{for all } g \in G.$$

If T is an invertible linear isomorphism, we say that ρ_1 and ρ_2 are *equivalent* (or *isomorphic*), denoted $\rho_1 \cong \rho_2$.

As we have shown that an inverse of an intertwining operator of representations of $k-$algebras is also intertwining, one can show the same statement for $G-$homomorphisms.

Definition 2.8 (Subrepresentation)

Let $\rho\colon G \to \mathrm{GL}(V)$ be a representation. A vector subspace $W \subseteq V$ is called a *G-invariant subspace* (or *subrepresentation*) if

$$\rho(g)w \in W \quad \text{for all } g \in G \text{ and all } w \in W.$$

Definition 2.9 (Irreducible representation)

We say a representation (V, ρ) is irreducible if V has no proper non-zero subrepresentations.

Definition 2.10 (Indecomposable representation)

We say a representation (V, ρ) is indecomposable if there doesn't exist a pair of subrepresentations U, W such that $V = U \oplus W$.

Note. It is essential to distinguish between *irreducibility* and *indecomposability*:

- A representation V is *irreducible* if it admits no proper non-zero subrepresentation.
- A representation V is *indecomposable* if it cannot be written as a direct sum of two non-zero subrepresentations, i.e., $V \neq W_1 \oplus W_2$ with $W_1, W_2 \neq \{0\}$.

Every irreducible representation is indecomposable, but the converse does not hold in general fields! As we shall see in the section of Maschke's Theorem of the next chapter, when the characteristic of the field divides the group order, there exist indecomposable representations that are not irreducible.

2.4 The Group Algebra $k[G]$

Now, we move to see the other way to define a group representation. We see it as a restriction of the definition of a group algebra representation on the basis elements which are basically the elements of the group G. But what is the group algebra?

Definition 2.11 (The Group Algebra $k[G]$)

Let G be a finite group and k a field. The *group algebra* $k[G]$ is the k-vector space with basis given by the elements of G:

$$k[G] = \left\{ \sum_{g \in G} a_g \, g \;\middle|\; a_g \in k \right\}.$$

The multiplication on $k[G]$ is defined by extending the group multiplication linearly:

$$\left(\sum_{g \in G} a_g \, g \right) \left(\sum_{h \in G} b_h \, h \right) = \sum_{g,h \in G} a_g b_h \, (gh) = \sum_{x \in G} \left(\sum_{gh = x} a_g b_h \right) x.$$

The vector space dimension of $k[G]$ over k is precisely the order of the group:

$$\dim_k(k[G]) = |G|.$$

Similar to the general definition 2.2 of a representation of an algebra, we define a representation of a group algebra.

Definition 2.12 (Representation of a group algebra $k[G]$)

Let G be a group. Let $k[G]$ be any group algebra over a field k. A representation of $k[G]$ is a pair (V, ρ), where V is a k-vector space and

$$\rho : A \to \mathrm{End}(V)$$

is a group algebra homomorphism.

The previous definition just says that we are trying to represent elements of an algebra with linear transformation or matrices. Observe that this gives rise to another way to see a group representation. The restriction of the function ρ to the elements of the group G yields exactly the definition of a group representation. Hence, we deduce that the definition of a representation of a group algebra is essentially the same data as a group representation.

Chapter 3

Semi-Simple Algebras

3.1 Definitions

Now, as we are done talking about the basics of representation theory, we get to the main goal of this exposition: introducing Semisimple algebras. The main claim we aim to prove in this chapter is that all semisimple algebras are product of matrix algebras. This characterization shows that semisimple algebras are the easiest to work with.

Definition 3.1 (Semisimple Algebra)

An algebra A is *semisimple* if when viewed as a representation of A is a direct sum of irreducible representations of A.

Definition 3.2

A completely reducible representation of an alegbra A is a direct sum of irreducible representations.

There are many ways to define semisimple algebras, and they are all equivalent. At the end, we see that all of these definitions are just stating properties that such an algebra has. We will prove that all of these properties are equivalent.

In fact, semisimple algebras are nice to work with since they are all isomorphic to a

direct sum of matrix algebra (Artin-Wedderburn). Moreover, we will see a characterization for an arbitrary group algebra over a field (with some restrictions on the characterstic of that field) to be semisimple thus nice, and this is the main title of this exposition which is Maschke's Theorem.

Now, we have a nice lemma about irreducible representations that makes a lot of sense if we can see that the kernel and the image of an intertwining operator are subrepresentations.

Lemma 3.3 (Schur's Lemma)

Let A be a $k-$algebra and V, W be two irreducible representations of A. Let $\varphi : V \to W$ be an intertwining operator. Then φ is either zero or an isomorphism.

Proof. Since $\ker \varphi$ is a subrepresentation of V then $\ker V = 0$ or $\ker V = V$.

1. Case 1: If $\ker V = 0$, then φ is injective. But also W is irreducible so $\operatorname{Img} \varphi = 0$ or $\operatorname{Img} \varphi = W$ by rank-nullity we get that $\operatorname{Img} \varphi$ must be W hence φ is an isomorphism.
2. Case 2: if $\ker V = V$ then φ is trivially the zero map.

and this ends the proof. ■

Lemma 3.4 (Schur's Lemma for algebraically closed fields)

Let A be any $k-$algebra, and V an irreducible representation of A. Then,

$$\operatorname{End}_A(V) \cong k \cdot id \cong k$$

Proof. Let $T \in \operatorname{End}_A(V)$ be any intertwining operator. Since k is algebraically closed then T has at least one eigenvalue in k. So, there is a non-zero vector $v \in V$ such that

$$T(v) = \lambda v$$

for some $\lambda \in k$. From this equation we get that $T - \lambda \operatorname{id}(v) = 0$. Since $T - \lambda \operatorname{id} \in \operatorname{End}_A(V)$ and V is irreducible then it is either zero or an isomorphism. But v is a non-zero vector in

$\ker\big(T - \lambda\,\mathrm{id}\big)$. Therefore $T - \lambda\,\mathrm{id} = 0$, where 0 is the zero map, and so $T = \lambda\,\mathrm{id}$. This completes the proof. ■

Before we go to the main theorems we define the regular and trivial representations

Definition 3.5 (Regular Representations)

Let A be any $k-$algebra. The regular representation of A is the vector space structure of A, with the action on itself given by the multiplication operation defined on A as an algebra. More precisely, the pair (A, ρ) is the regular representation of an algebra A, where $\rho : A \to \mathrm{End}(A)$ is given by the multiplication operation where:

$$a \cdot b = \rho(a)(b) = ab$$

for all $a, b \in A$

Definition 3.6 (Trivial Representation)

Let $A = \mathrm{Mat}_d(\mathbb{R})$ be an algebra. The *trivial representation* of A is the vector space $V = k^{\oplus d}$, where the action of A is given by regular matrix multiplication.

Definition 3.7 (Radical)

The radical of a finite dimensional algebra A denoted by $\mathrm{Rad}(A)$ is an ideal of A containing all elements that are mapped to the zero map in all irreducible representations of A. More precisely,

$$\mathrm{Rad}(A) = \bigcap_i \ker(\rho_i)$$

where (V_i, ρ_i) are the irreducible representations of A.

We will then prove that the radical of A is a two sided ideal

Proof. Let (V, ρ) be any irreducible representation of A. Pick any element $a \in A$ and an

element $r \in \text{Rad}(A)$. Then $\rho(r)(v) = 0$ for any $v \in V$. Since ρ is a homomorphism of algebras, we get that

$$\rho(ar)(v) = \rho(a)(\rho(r)(v)) = \rho(a)(0) = 0$$

So, $ar \in \text{Rad}(A)$. For the right side, since $r \in \text{Rad}(A)$ then:

$$\rho(ra)(v) = \rho(r)(\rho(a)(v)) = 0$$

Thus, $ra \in \text{Rad}(A)$, showing that $\text{Rad}(A)$ is a two sided ideal ■

3.2 Artin-Wedderburn Theorem

Theorem 3.8 (Equivalences to semisimplicity)

Let A be a finite dimensional algebra. **TFAE**:

1. $\text{Rad}(A) = 0$.
2. $\sum_{i=1}^{r} \dim(V_i)^2 = \dim(A)$,
3. $A \cong \bigoplus_{i=1}^{r} \text{Mat}_{d_i}(k)$, where d_i is the dimension of the irreducible representation V_i of A.
4. Any finite dimensional representation of A is completely reducible.
5. A is a semi-simple algebra

Proof. ■

Theorem 3.9 (The irreducible representations of a matrix algebra)

Let $A = \mathrm{Mat}_d(k)$ be a matrix algebra. The only irreducible representation of A is the trivial representation $V = k^{\oplus d}$. Moreover, any finite dimensional representation W of A is a direct sum of $k^{\oplus d}$ i.e. $W = V^{\oplus m}$ for some m.

The previous theorem completely describes the finite dimensional representations of any semisimple algebra (as they are direct sum of matrix algebras) and states that there is only one irreducible one. In the next section, we will show that more generally for any finite dimensional algebra we have finitely many irreducible representations, and it turns out that the semisimple algebras are the nice case when the number of the irreducible representations is just one.

3.3 Density Theorem

Before we introduce the density theorem we define the notation of a direct sum of functions denoted by $\bigoplus_{i=1}^{r} \rho_i$.

Definition 3.10

For any set A and a collection of r many sets B_i, where $1 \leq i \leq r$. Define r many functions $\phi_i : A \to B_i$, then the *direct sum* of ϕ_i is the function $\bigoplus_{i=1}^{r} \phi_i$ where:

$$\bigoplus_{i=1}^{r} \phi_i : A \to \bigoplus_{i=1}^{r} B_i,$$

given by

$$a \mapsto (\phi_1(a), \phi_2(a), \ldots, \phi_r(a))$$

or in a more compact notation $a \mapsto \bigoplus_{i=1}^{r} \phi_i(a)$ for every $a \in A$.

Theorem 3.11 (Density Theorem)

Let A be a finite dimensional algebra, and $V_1, V_2, \ldots, V_r$ be the irreducible representations of A, then

1. For any i we have that $\rho_i : A \to \mathrm{End}(V_i)$ is surjective
2. The direct sum map $\bigoplus_{i=1}^m \rho_i : A \to \bigoplus_{i=1}^m \mathrm{End}(V_i)$ is surjective

So, from definition 3.3 we can see that the image of A in $\bigoplus_{i=1}^m \mathrm{End}(V_i)$ is a subset of the direct sum of the images of each ρ_i. To be more precise, we define the following sets for our proof:

1. The image of A in $\bigoplus_{i=1}^m \mathrm{End}(V_i)$ is the set $B = \left\{ \bigoplus_{i=1}^m \rho_i(a) \mid a \in A \right\}$.
2. The sets $B_i = \mathrm{Img}(\rho_i)$. Their direct sum $\bigoplus_{i=1}^m B_i = \left\{ \bigoplus_{i=1}^m \rho_i(a_i) \mid a_i \in A \right\}$

Note. Those two sets B and $\bigoplus_i B_i$ seem to be equal at the first glance, but in fact $B \subseteq \bigoplus_{i=1}^m B_i$. This is because the tuples in B have the same a as an input for each ρ_i, while in the direct sum $\bigoplus_{i=1}^m B_i$ it is not necessarily the case that the input to each ρ_i is the same. Now, what we will try to prove and what actually the theorem is telling us is that no matter what tuple of matrices you choose in $\bigoplus_{i=1}^m \mathrm{End}(V_i)$, or for any tuple with any combination of inputs to ρ_i, we will be able to find an input element $a \in A$ such that the tuple we chose is exactly $\bigoplus_{i=1}^m \rho_i(a)$ (note that it is the same a!). This is what surjectivity actually mean here, we pick a tuple in $\bigoplus_{i=1}^m \mathrm{End}(V_i)$, which turns out to be equal to the direct sum $\bigoplus_i B_i$, and we want to show that this tuple is in the image of A under the function $\bigoplus_{i=1}^m \rho_i$.

We introduce the following two lemmas in our proof

Lemma 3.12

Let $V = \bigoplus_{i=1}^{m} V_i^{\oplus n_i}$ where V_i are pairwise non-isomorphic representations. Then any subrepresentation of V is of the form $W = \bigoplus_{i=1}^{m} V_i^{\oplus m_i}$ where $m_i \leq n_i$ for every i, and the inclusion map $\phi : W \hookrightarrow V$ is given by a direct sum of inclusions of each summand $\phi_i : V_i^{\oplus n_i} \hookrightarrow V_i^{\oplus m_i}$, which are $n_i \times m_i$ matrices.

The intuition behind this lemma comes from Schur's lemma.

To be continued

Lemma 3.13

Let V be an irreducible representation of A and consider any set of linearly independent vectors $v_1, v_2, \ldots, v_r \in V$. Then for any vectors $w_1, w_2, \ldots, w_r \in V$, there is $a \in A$ such that $av_i = w_i$ for every i.

This lemma basically says that any set of vectors are mapped via some element in A to a linearly independent set of vectors.

Proof. Let V be an irreducible representation of dimension n and $v_1, v_2, \ldots v_n \in V$ be a linearly independent set. Consider the map $\phi : A \to V^{\oplus n}$ given by

$$a \mapsto (av_1, av_2, \ldots, av_n)$$

. Assume for the sake of contradiction that there are vectors $w_1, w_2, \ldots, w_n$ such that for any $a \in A$ we can't have $av_i = w_i$. Therefore, vectors $w_i \notin \mathrm{Img}(\phi)$. So, the image of ϕ is a proper subrepresentation of $V^{\oplus n}$. By Lemma 3.3 $\mathrm{Img}(\phi)$ is of the form $V^{\oplus d}$ where $d < n$. Since the map of inclusion $\mathrm{Img}(\phi) \to V^{\oplus n}$ is given by some $d \times n$ matrix X, then by rank-nullity we can find a nonzero vector of scalars $c = [c_1, c_2, \ldots, c_n]$ such that $X\mathbf{c} = 0$.

Now, consider the sum

$$\sum_i c_i v_i = [c_1, c_2, \ldots, c_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \tag{3.1}$$

Note that $\phi(1) = (v_1, v_2, \ldots, v_n)$, so $(v_1, v_2, \ldots, v_n) \in \text{Img}(\phi)$. Thus

$$[v_1, v_2, \ldots, v_n]^T = [q_1, q_2, \ldots, q_d]X$$

Substituting for $\mathbf{v}^T$ in (3.1) we get:

$$\sum_i c_i v_i = [q_1, q_2, \ldots q_d]X[c_1, c_2, \ldots, c_n]^T = \mathbf{0} \times [q_1, q_2, \ldots, q_d] = 0$$

contradicting that v_i are linearly independent. This completes the proof. ■

Lemma 3.14

Let A be a finite dimensional algebra. Let (V, ρ) be a representation of A. Consider Mat(V) as a representation of A with the action given by matrix multiplication:

$$a \cdot M = \rho(a) \cdot M.$$

Then the following is true:

1. The map $\rho : A \to \text{Mat(V)}$ is an intertwining operator, where A is the regular representation.
2. We have the following isomorphism of representations $\text{Mat}(V) \cong V^{\oplus d}$, where d is the dimension of V.

Proof. First we prove (1). Let A and $\text{Mat}(V)$ be as before. To show that ρ is intertwining

we will show that for any $a, b \in A$ we have that $\rho(ab) = a \cdot \rho(b)$, where ab is a product in A and $\cdot$ is the action of a on $\rho(b) \in \mathrm{Mat}(V)$. By definition of V as $A-$module we have that $\rho(ab) = \rho(a) \cdot \rho(b)$. By definition of the action of A, where $a \cdot M = \rho(a) \cdot M$ for all $a \in A$ and $M \in \mathrm{Mat}(V)$, we get the desired equality

$$\rho(ab) = \rho(a) \cdot \rho(b) = a \cdot \rho(b).$$

For 2, consider the map $\phi : \mathrm{Mat}(V) \to V^{\oplus d}$ given by mapping every matrix in $\mathrm{Mat}(V)$ to the vector in $V^{\oplus d}$ with the $i-$th entry representing the $i-$th column in the matrix. That is

$$\begin{pmatrix} m_{11} & m_{12} & \dots & m_{1d} \\ m_{21} & m_{22} & \dots & m_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ m_{d1} & m_{d2} & \dots & m_{dd} \end{pmatrix} \mapsto \left(\begin{bmatrix} m_{11} \\ m_{21} \\ \vdots \\ m_{d1} \end{bmatrix}, \begin{bmatrix} m_{12} \\ m_{22} \\ \vdots \\ m_{d2} \end{bmatrix}, \dots, \begin{bmatrix} m_{1d} \\ m_{2d} \\ \vdots \\ m_{dd} \end{bmatrix} \right)$$

This map is clearly surjective. For injectivity suppose for any $M_1, M_2 \in \mathrm{Mat}(V)$ we have $\phi(M_1) = \phi(M_2)$, then the columns of the two matrices are equal so $M_1 = M_2$. It remains to show that ϕ is intertwining that is $\phi(a \cdot M) = a \cdot \phi(M)$. By definition of action of a on M we get

$$\phi(a \cdot M) = \phi(\rho(a) \cdot M)$$

By definition of matrix multiplication, it works on columns independently. Let

$$\rho(a) = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1d} \\ a_{21} & a_{22} & \dots & a_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ a_{d1} & a_{d2} & \dots & a_{dd} \end{pmatrix}, \quad M = \begin{pmatrix} m_{11} & m_{12} & \dots & m_{1d} \\ m_{21} & m_{22} & \dots & m_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ m_{d1} & m_{d2} & \dots & m_{dd} \end{pmatrix}$$

We can write $\rho(a)$ and M in a more compact notation $a = \begin{pmatrix} \mathbf{a_1} \\ \mathbf{a_2} \\ \vdots \\ \mathbf{a_d} \end{pmatrix}$, where $\mathbf{a_i}$ is the $i-$th row.

$$M = \begin{pmatrix} \mathbf{c_1} & \mathbf{c_2} & \dots & \mathbf{c_d} \end{pmatrix}, \text{ where } \mathbf{c_j} \text{ is the } j-\text{th column.}$$

So, the product $\rho(a) \cdot M$ is

$$\begin{pmatrix} \mathbf{a_1c_1} & \mathbf{a_1c_2} & \dots & \mathbf{a_1c_d} \\ \mathbf{a_2c_1} & \mathbf{a_2c_2} & \dots & \mathbf{a_2c_d} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a_dc_1} & \mathbf{a_dc_2} & \dots & \mathbf{a_dc_d} \end{pmatrix}$$

The $i-$th column of the product could be seen as $(d \times 1)-$matrix resulting from multiplying the matrix $a_{d\times d}$ with the $i-$th column of the matrix $M_{d\times 1}$. So, the $i-$th column of the product is exactly the $i-$th entry of the vector $a \cdot \phi(M)$, where a acts on this vector by matrix multiplication with each entry as follows:

$$a \cdot \phi(M) = \left(a \cdot \begin{bmatrix} m_{11} \\ m_{21} \\ \vdots \\ m_{d1} \end{bmatrix}, a \cdot \begin{bmatrix} m_{12} \\ m_{22} \\ \vdots \\ m_{d2} \end{bmatrix}, \dots, a \cdot \begin{bmatrix} m_{1d} \\ m_{2d} \\ \vdots \\ m_{dd} \end{bmatrix} \right)$$

Therefore $\phi(a \cdot M) = a \cdot \phi(M)$ as desired. ■

Now, we are ready to prove the density theorem

Proof. 1. Since V_i is a representation of A, then $\text{End}(V_i) \cong \text{Mat}(V_i) \cong V_i^{\oplus d_i}$, where d_i is the dimension of V_i. Since $\text{Img}(\rho_i)$ is a subrepresentation of $\text{End}(V_i)$ then by lemma 3.3, $\text{Img}(\rho_i)$ is of the form $V^{\oplus m}$ for some $m \leq d$. We will show that $m = d$. Let

$v_1, v_2, \ldots, v_d$ be a basis of $V^{\oplus d}$. By linear algebra we know that the columns of the any matrix in $\mathrm{Mat}(V)$ is just the images of the basis vector. So, via ρ_i and the isomorphism $\mathrm{End}(V_i) \cong \mathrm{Mat}(V_i) \cong V^{\oplus d}$, we get $a \mapsto (av_1, av_2, \ldots, av_n)$. This is exactly the image of A in $\mathrm{End}(V_i)$, because the matrix $\rho(a)$ is completely determined by where it sends the basis vectors of V. Pick any vector $(w_1, w_2, \ldots, w_d) \in V^{\oplus d}$, where each $w_i \in V$. By lemma 3.3, there is $a \in A$ such that $(w_1, w_2, \ldots, w_d) = (av_1, av_2, \ldots, av_d)$. Therefore, $(w_1, w_2, \ldots, w_d) \in \mathrm{Img}(\rho_i) = V^{\oplus m}$, then $V^{\oplus d} \subseteq V^{\oplus m}$, and $m = d$ yielding surjectivity as desired.

2. For the case of $r \geq 2$, we get that the image of the direct sum $\bigoplus_{i=1}^{r} \rho_i$ is a subrepresentation of $\bigoplus_{i=1}^{r} \mathrm{End}(V_i) \cong \bigoplus_{i=1}^{r} V_i^{\oplus d_i}$. By lemma 3.3, the image is of the form $\bigoplus_{i=1}^{r} V_i^{\oplus m_i}$, and the inclusion is a direct sum of the inclusions of each summand. By (1), we have shown that the inclusion of each summand is surjective. Therefore, the map $\bigoplus_{i=1}^{r} \rho_i$ is surjective.

■

Corollary 3.15 (Finitely many irreps)

Let A be any finite dimensional algebra, then A has finitely many finite dimensional irreducible representations V_i. Moreover, we have the following isomorphism of algebras:

$$A \Big/ \mathrm{Rad}(A) \cong \bigoplus_i \mathrm{End}(V_i)$$

Proof. Let A be as above. Consider V an irreducible representation of A. Then for any non-zero $v \in V$ we have that Av is a subrepresentation (a subspace closed under the action of A). Since $Av \neq 0$ and V is irreducible then $Av = V$. Since A is finite dimensional then V is finite dimensional. Since the irreducible representations we considered in Theorem 3.3 are pairwise non isomorphic, we get that $\sum_{i=1}^{r} \dim(V)^2 = \dim(\bigoplus_{i=1}^{r} \mathrm{End}(V_i))$. By surjectivity of the map ρ in Theorem 3.3, and rank-nullity theorem we get that $\sum_{i=1}^{r} \dim(V_i)^2 \leq \dim(A)$.

Since $\dim(V_i) \geq 1$ then $r \leq \sum_{i=1}^{r} \dim(V_i)^2 \leq \dim(A)$, where r is the number of the irreducible representations of A, yielding that they are finite as desired.

For the moreover part, clearly $\mathrm{Rad}(A) \subseteq \ker(\rho)$. For the other inclusion, let $a \in A$ be an element of $\ker(\rho)$. Then $\rho(a) = (0, 0, \ldots, 0)$, where each entry is $\rho_i(a)$. This shows that a is mapped to the zero map in every irreducible representation. Thus $a \in \mathrm{Rad}(A)$. So, $\ker(\rho) \subseteq \mathrm{Rad}(A)$. By the universal property of quotients the map ρ factors through $\mathrm{Img}(\rho)$. Since ρ is surjective then $\mathrm{Img}(\rho) = \bigoplus_i \mathrm{End}(V_i)$ showing the desired isomorphism. ■

3.4 Maschke's Theorem

Theorem 3.16 (Maschke Theorem)

Let G be a finite group, and k be a field where $\mathrm{char}(k) \nmid |G|$. Let (V_i, ρ_i) be the irreducible representations of G for $1 \leq i \leq r$, then

1. $k[G]$ is semi-simple.
2. $\phi : k[G] \to \bigoplus_{i=1}^{r} \mathrm{End}(V_i)$ is an isomorphism of algebras.

First, we introduce the definition of an idempotent map.

Definition 3.17

Let V be a vector space. Let $P : V \to V$ be a linear map. We say P is idempotent if $P^2 = P$ for all $v \in V$.

Corollary 3.18

Let P be as before, then P is the identity map on its image.

Proof. Since P is idempotent then $P^2 = P$ for any $v \in V$. Consider $u \in \mathrm{img}(P)$, then

$u = P(v)$ for some $v \in V$. Thus,

$$P(u) = P(P(v)) = P(v) = u.$$

■

We use the following lemmas in the proof.

Lemma 3.19

For any idempotent map $P : V \to V$, we have that

$$V = \text{Img}(P) \oplus \ker(P).$$

as vector spaces.

Lemma 3.20

Let U, W be any subspaces of V. Suppose $V = U \oplus W$, then $U = \text{Img}(P)$ and $W = \ker(P)$ for some idempotent map P.

This lemma shows that any decomposition of V into a direct sum is given by some idempotent map in the background.

Proof. Consider P the projection along W onto U that is for any $v = (u, w) \in V$, where $u \in U$ and $w \in W$, we have that $P((u, w)) = (u, 0)$. then $P^2 = P$ for every $v \in V$. By definition of P we get $\text{Img}(P) \cong U$ and $\ker(P) \cong W$ because a vector v in the kernel is mapped to zero, which means its u coordinate is 0 i.e. of the form $(0, w)$. Clearly, P is idempotent, because

$$P(P((u, w))) = P((u, 0)) = (u, 0), \tag{3.2}$$

Observe that the first equality in (3.2) holds because $\text{Img}(P) = U$, and the second holds because $P\,|_U = \text{id}$.

■

Now, we are ready to prove Maschke's Theorem.

Proof. First, suppose (V, ρ) is a representation of G, and $W \subset V$ is a subrepresentation of V. We will show that there is W' a subrepresentation of V such that $V = W \oplus W'$ as representations. Consider the projection $P : V \to W$ onto W. So, $\text{Img}(P) = W$, and $P\,|_W = \text{id}$. We will construct a subrepresentation W' such that $V = W \oplus W'$.

Step 1: $\text{Img}(\overline{P}) = W$

Consider the average map $\overline{P} : V \to V$ defined as

$$\overline{P} := \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g^{-1})$$

Let $\ker(\overline{P}) = W'$. Pick any $v \in V$, then

$$\begin{aligned}
\overline{P}(v) &= \frac{1}{|G|} \sum_{g \in G} \rho(g)\big(P\big(\rho(g^{-1})(v)\big)\big) \\
&= \frac{1}{|G|} \sum_{g \in G} \rho(g)\big(P(v')\big) \\
&= \frac{1}{|G|} \sum_{g \in G} \rho(g)(w) \\
&= \frac{1}{|G|} \sum_{g \in G} w' = \frac{1}{|G|} \times |G| \cdot w' = w' \in W.
\end{aligned}$$

So, $\mathrm{Img}(\overline{P}) \subseteq W$. To show the other inclusion, we will show that $\overline{P}$ is the identity on W. Pick any $w \in W$

$$\begin{aligned}\overline{P}(w) &= \frac{1}{|G|}\sum_{g\in G}\rho(g)\big(P\big(\rho(g^{-1})(w)\big)\big)\\ &= \frac{1}{|G|}\sum_{g\in G}\rho(g)\big(\rho(g^{-1}w)\big)\\ &= \frac{1}{|G|}\sum_{g\in G} w = \frac{1}{|G|}\times|G|\cdot w = w.\end{aligned}$$

Thus, $\mathrm{Img}(\overline{P}) = W$. Pick any $v \in V$, the image $\overline{P}(v) = w$ for some $w \in W$, then

$$\overline{P}(\overline{P}(v)) = \overline{P}(w) = w = P(v)$$

So, $\overline{P}$ is idempotent. Hence, $V = \mathrm{Img}(\overline{P}) \oplus \ker(\overline{P}) = W \oplus W'$ as vector spaces. We know W is a subrepresentation. It remains to show that $W' = \ker(\overline{P})$ is a subrepresentation.

Step 2: $\overline{P}$ is intertwining

Pick any $h \in G$ and any $v \in V$, we will show that $\overline{P}(\rho(h)v) = \rho(h)\overline{P}(v)$.

$$\begin{aligned}\overline{P}(\rho(h)v) &= \frac{1}{|G|}\sum_{g\in G}\rho(g)\big(P\big(\rho(g^{-1})(\rho(h)(v))\big)\big)\\ &= \frac{1}{|G|}\sum_{g\in G}\rho(g)\big(P\big(\rho(g^{-1}h)(v)\big)\big)\end{aligned}$$

Let $l^{-1} = g^{-1}h$. So, $l = (g^{-1}h)^{-1} = h^{-1}g$ Then $g = hl$. Substitute in the last equation

$$\begin{aligned}
\overline{P}(\rho(h)v) &= \frac{1}{|G|}\sum_{l\in G}\rho(hl)\big(P\big(\rho(l^{-1})(v)\big)\big) \\
&= \frac{1}{|G|}\sum_{l\in G}\rho(h)\rho(l)\big(P\big(\rho(l^{-1})(v)\big)\big) \\
&= \rho(h)\frac{1}{|G|}\sum_{l\in G}\rho(h)\rho(l)\big(P\big(\rho(l^{-1})(v)\big)\big) \\
&= \rho(h)\overline{P}(v).
\end{aligned}$$

as desired. Since $\overline{P}$ is intertwining, then $\ker(\overline{P})$ is a subrepresentation. Therefore, $V = W \oplus W'$ as representations.

Step 3

We have shown that if V is a representation and $W \subset V$ is a subrepresentation then $V = W \oplus W'$ for some subrepresentation W'. Now, we will show by induction on $\dim(V)$, under the same hypotheses, that V is a direct sum of irreducible representations. If $\dim(V) = 1$ then it is immediate that V is irreducible. Let $n \geq 2$. Suppose that for any representation V' of $\dim \leq n$ is a direct sum of irreducible representations. Let V be a representation with $\dim(V) = n + 1$. By **Step 2**, $V = W \oplus W'$, where $\dim(W) \leq n$, and $\dim(W') \leq n$. By the induction hypothesis W, W' are direct sums of irreducible representations, showing the claim is true. Since any representation V of G is completely reducible, then $k[G]$ is semi-simple as desired. ■

Note. Observe that the proof of Maschke's theorem shows that, under the hypothesis of the theorem, $k[G]$ is semi-simple by showing that any representation of $k[G]$ is completely reducible (since they are equivalent as in theorem 3.2). Therefore, any indecomposable representation is irreducible. One can easily show that this fails in general. That is there are indecomposable representations of $k[G]$ that are not irreducible, if $\mathrm{char}(k)$ divides $|G|$.

Now, we give an example of an indecomposable representation of G which is reducible. This happens if $\mathrm{char}(k) \mid |G|$.

Corollary 3.21 (The sum of squares formula)

Let V_i be as above, then

$$|G| = \sum_{i=1}^{r} \dim(V_i)^2.$$

Proof. The proof is immediate from Theorem 3.2 (2), and the fact that $\dim(k[G]) = |G|$. ■